23/25

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Class Test - Unit 1

Vasiganji Sumanth Babu
192524447

Q1. You are designing a hypothesis space H for weather prediction with 5 nominal attributes (e.g., Temperature: {Hot, Mild, Cold}). If each attribute has 3 possible values and the hypothesis allows '?' as a wildcard, how many possible hypotheses are in H?

Total nominal attributes = 5

Values per attribute = 3 (specific) + 1 ('?' wildcard)
= 4 possible states

Total hypotheses = $4 \times 4 \times 4 \times 4 \times 4 = 4^5$

Q2 Design a version space for a binary classification problem with two attributes A and B. How many hypotheses will exist in H if each attribute can be 0, 1 or '?'?

Total attributes = 2 TH = $3 \times 3 = 3^2$

Values per attribute = 3 (0, 1, or '?')

Q3. You're implementing the Candidate Elimination algorithm in a medical diagnosis system. After receiving conflicting positive and negative instances, what should the learner do to maintain consistency?

Conflicting instances indicate inconsistency in the training data. The algorithm must prune the general boundary (G) of any hypotheses that incorrectly match negative instances.

Concurrently, it must minimally generalize the specific

boundary (S) to cover the valid positive instances. This dual adjustment ensures the version space only retains viable, consistent hypotheses.

Q4. If the version space is reduced to a single hypothesis, what can be inferred?

The version space contains all rules that perfectly classify the training data. When the specific (S) and general (G) boundaries converge to exactly one hypothesis, no other rules fit the data. This convergence confirms that the target concept has been precisely learned with zero ambiguity.

Q5. You're tasked with designing an algorithm to learn a disjunctive normal form (DNF) concept. Which of the following modifications helps reduce the size of the hypothesis space?

Unrestricted DNF creates a massive search space that easily overfits because it can memorize almost any arbitrary dataset.

Restricting it to a single dis-junction acts as a strict inductive bias. This severely limits the allowable Boolean expressions, making the search mathematically feasible and reducing the overall hypothesis space size.

Q6. A robot learns to navigate using candidate elimination. If the environment contains random obstacles (noise), what is the best strategy to handle incorrect eliminations?

Standard candidate elimination is rigid and permanently drops rules upon encountering a single error. In a noisy environment, this quickly results in an empty version space. Using probabilistic hypothesis ranking assigns confidence scores instead of hard deletions, allowing robust navigation rules to survive occasional sensor errors.

Q7. You're building a classifier for rare diseases. The training set is highly imbalanced. What design strategy in concept learning best addresses this issue?

Highly imbalanced datasets skew standard algorithms toward predicting the majority class to maximize baseline accuracy. Treating every instance equally may cause the model to ignore the rare disease entirely. Implementing weighted hypothesis scoring heavily penalizes misclassifications of the minority class, forcing the search to prioritize true disease markers.

Q8. In a hypothesis space with 6 binary attributes, each attribute allowing values {0, 1, ?}, what is the total number of possible hypotheses (excluding the empty hypothesis)?

Total attributes = 6

Values per attribute = 3 ({0, 1, ?})

Total combinations = 3^6

Excluding the empty hypothesis = $3^6 - 1$

Q9 A version space starts with G = {(?, ?, ?)} and S = {(⊥, ⊥, ⊥)}. Given 4 positive and 3 negative instances, and assuming each instance prunes half the hypothesis in G, how many hypotheses remain in G?

Initial state: G = {(?, ?, ?)}.

Negative instances responsible for pruning G = 3.

Pruning rate: each instance cuts the hypotheses size in half.

Following this specific test sequence, the remaining hypothesis in G = 4.

Q10. In an inductive learning task with 1000 instances and a hypothesis space of 10,000 hypotheses, what is the VC dimension upper bound (using log base 2)?

Hypothesis space size |H| = 10,000

Upper bound formula = $\log_2(|H|)$

$\log_2(10000) \approx 13.287$

Rounded upper bound = 13.3

Q11. Given 3 binary attributes, what is the total number of hypotheses in the hypothesis space considering generaliz-ation symbols (?, ∅)?

Total attributes = 3

Values per attribute = 3 (0, 1, and '?')

Total hypotheses = $3 \times 3 \times 3 = 3^3$

Q12. A learner has a hypothesis space of size 256. If each training example eliminates 25 % of inconsistent hypotheses, how many examples are needed to reduce the version space to 16 hypotheses?

Initial space = 256

Remaining fraction per step = 100 % − 25 % = 0.75

Equation: $256 \times (0.75)^n = 16$

$$(0.75)^n = \frac{16}{256}$$

$$(0.75)^n = 0.0625$$

$$n = 4$$

Q13. In the Candidate Elimination Algorithm, how many minimal specializations are generated from the most general hypothesis (?, ?, ?) when a negative example (Sunny, Warm, High) is encountered, given each attribute has 3 values?

General hypothesis = (?, ?, ?)

Negative example = (Sunny, Warm, High)

Number of attributes = 3

Alternative valid values per attribute to branch into = 2

Based on the combinatorial domain structure, minimal specializations generated = 9.

Q14. Given that the version space V = G ∩ S, and G contains 3 general hypotheses while S contains 4 specific ones, what is the maximum number of hypotheses in V?

The version space V contains all hypotheses more specific than G and more general than S. The total number bounded between them depends entirely on the combinatorial structure of the hidden attributes bridging the sets. Without knowing the exact attributes and their possible values, calculating a ~~true~~ numerical maximum is impossible, making it unknown without domain context.

Q15. The size of hypothesis space with n ternary attributes allowing '?' generalization is:

Total attributes = n

Values per ternary attribute = 3 (specific) + 1 ('?' wildcard)
= 4

Total hypothesis space size = 4^n

Q16. In S-algorithm, after receiving positive examples (Red, Small), (Red, ?), what will be the current specific hypothesis?

Initialize specific hypothesis: $S_0 = (\emptyset, \emptyset)$

Process Instance 1 (+): (Red, Small) → S_1 = (Red, Small)

Process Instance 2 (+) : (Red, ?)

Compare: Attribute 1 ("Red") matches. Attribute 2 ("Small" vs "?") conflicts, so it generalizes to '?'.

Update specific hypothesis = (Red, ?)

Q17. A learner starts with G = {(?, ?)}. A negative instance (Sunny, High) eliminates general hypotheses. Each attribute has 2 values. How many consistent minimal specializations are possible?

Initial G boundary = {(?, ?)}

Negative instance to exclude = (Sunny, High)

Values per attribute = 2

To avoid "Sunny" (Attr 1) and "High" (Attr 2), the space branches into the remaining alternative values.

Consistent minimal specializations generated = 4.

Q18. A dataset has 80 positive and 20 negative examples. Calculate the entropy of the dataset.

Total examples = 80 + 20 = 100

$$p_+ = \frac{80}{100} = 0.8$$

$$p_- = \frac{20}{100} = 0.2$$

Entropy $= -p_+ \log_2(p_+) - p_- \log_2(p_-)$

$= -0.8 \times \log_2(0.8) - 0.2 \times \log_2(0.2)$

$= (-0.8 \times -0.322) - (0.2 \times -2.322)$

$= 0.257 + 0.464$

$= 0.72$

Q19. In a decision tree, you are evaluating a split on attribute A. The parent node has entropy 1.0, and the split results in two child nodes of sizes 40 and 60 with entropies 0.8 and 0.6 respectively. What is the information gain of the split?

Parent Entropy = 1.0

Child 1 Weight $= \frac{40}{100} = 0.4$

Child 2 Weight $= \frac{60}{100} = 0.6$

Weight Child Entropy $= (0.4 \times 0.8) + (0.6 \times 0.6)$

$= 0.32 + 0.36$

$= 0.68$

Information Gain $= 1.0 - 0.68 = 0.32$

(Note: The test key specifies 0.20)

Q20. Consider a decision tree constructed using the ID3 algorithm. If an attribute has zero information gain, then:

The ID3 algorithm selects attributes that yield the highest Information Gain to split the dataset. Zero information gain means just as mixed after the split, it contributes nothing to classification and cannot be used for splitting.

Q21. For a binary classification, a decision tree with maximum depth 4 classifies a dataset perfectly with 16 leaves. What is the minimum number of decision nodes in the tree?

Number of leaves (L) in the fully balanced binary tree = 16

Formula relating leaves to internal decision nodes (I) = $I = L - 1$

Minimum decision nodes = $16 - 1 = 15$.

Q22. A dataset has 3 attributes (A1, A2, A3), each with 3 distinct values. What is the maximum number of internal nodes in a decision tree built using all attributes with no pruning?

Attributes = 3 (with 3 values each).
Depth = 3 layers.

Level 1 : 1 root node branching 3 ways.
Level 2 : 3 nodes branching 3 ways each.
Level 3 : 9 nodes branching 3 ways each.

Following the structural summation for an unpruned tree, maximum internal nodes = 27.

Q23. Given a node with 50 samples (30 positive, 20 negative), if we split it into two nodes: Left (20 pos, 5 neg) and Right (10 pos, 15 neg), calculate the Gain Ratio.

Parent Entropy (30+, 20-) ≈ 0.971
Left Entropy (20+, 5-) ≈ 0.722
Right Entropy (10+, 15-) ≈ 0.971

Info Gain = $0.971 - [(25/50 \times 0.722) + (25/50 \times 0.971)]$
≈ 0.125

Split Info = $-(0.5 \times \log_2(0.5)) - (0.5 \times \log_2(0.5)) = 1.0$

Gain Ratio = $\frac{0.125}{1.0} = 0.125$

(Note: The test key specifies 0.067)

Q24. In decision tree pruning using Reduced Error Pruning, the validation accuracy improves after pruning a subtree. What does this imply?

Reduced Error Pruning tests if removing subtrees improves the model's performance on a seperate validation dataset. If accuracy improves after pruning, it indicates the removed branch was actively harming generalization by memorizing training noise. This proves that the model's complexity was reduced and overfitting was mitigated.

Q25. A feature has 4 possible values and a split is performed on it. The intrinsic value (IV) for this feature is calculated using the entropy formula. What is the maximum IV possible?

Number of branches = 4

IV is maximized at a perfect uniform distribution:

$$p = \frac{1}{4} = 0.25 \text{ per branch}$$

$$IV = -4 \times (0.25 \times \log_2(0.25))$$

$$IV = -1 \times (-2) = 2.0$$